



INSTITUT NATIONAL
DES SCIENCES
APPLIQUÉES
CENTRE VAL DE LOIRE

Rapport Déploiement de VM Automatisé

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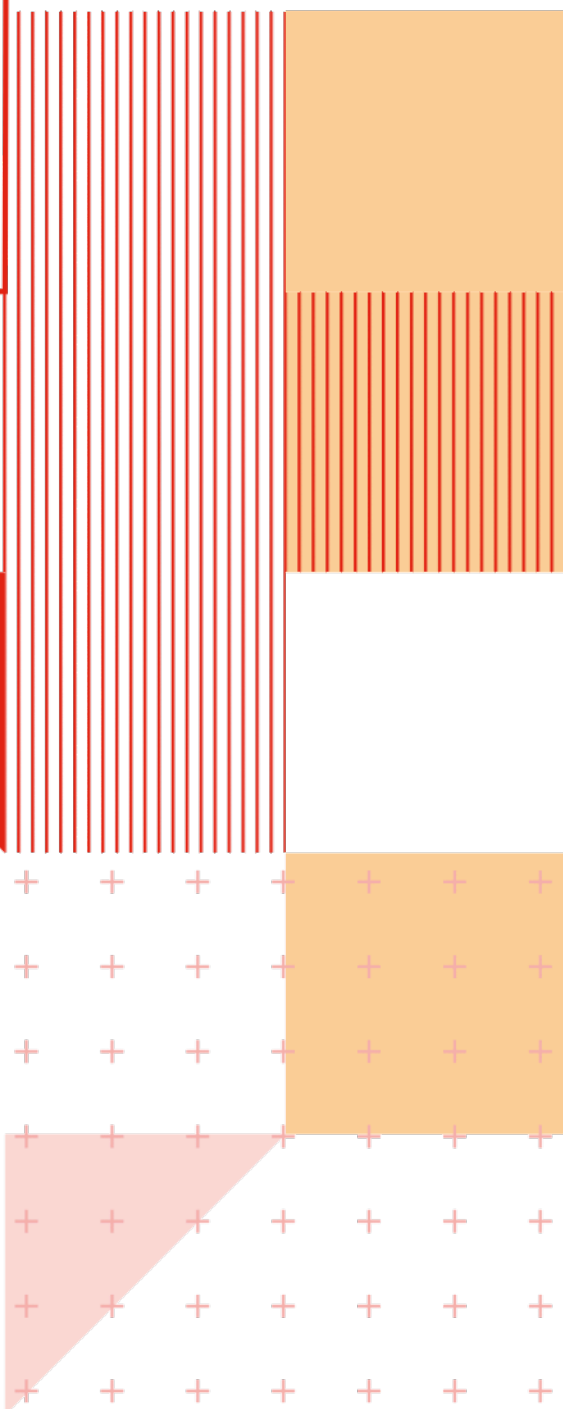


Table des matières

1. Premier déploiement avec TerraForm	3
1.1. Installation de Terraform	3
1.2. Création d'un user + token	3
1.3. Génération des fichiers de configuration Terraform	4
1.4. Terraform init et plan	5
1.5. Terraform apply	8
1.6. Terraform destroy	10
2. Déploiement avec adresse MAC fixée	12
2.1. DNSMasq to KEA & Unbound	12
2.2. Génération d'une adresse MAC statique	14
2.3. Attribution d'une IP statique à l'adresse MAC de la VM clonée	15
2.4. Terraform plan	16
2.5. Terraform apply	18
2.6. Terraform destroy	21
3. Implémentation de Cloud-init	22
3.1. Activation snippets dans local	22
3.2. Création d'un yml pour Cloud-init	22
3.3. Gestion de l'envoi du yml dans local:snippets	23
3.4. Dernières modifications du main.tf	23
3.5. Terraform plan	25
3.6. Terraform apply	27
3.7. Terraform destroy	31
4. Cleanup de l'espace de travail	34

1. Premier déploiement avec TerraForm

1.1. Installation de Terraform

Le déploiement de VM automatique se fera avec Terraform, qu'il convient d'installer. Après avoir récupéré le binaire depuis [hashicorp.com](https://releases.hashicorp.com/terraform/), il faut déplacer l'exécutable dans le dossier adéquat, et vérifier le bon fonctionnement de Terraform (test version):

```
user@debian:~/Téléchargements$ unzip terraform_1.16.2_linux_amd64.zip
Archive:  terraform_1.16.2_linux_amd64.zip
replace LICENSE.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
  inflating: LICENSE.txt
  inflating: terraform
user@debian:~/Téléchargements$ sudo mv terraform /usr/local/bin/
user@debian:~/Téléchargements$ terraform version
Terraform v1.16.2
on linux_amd64
```

1.2. Création d'un user + token

Après avoir créé, dans le CLI du proxmox, un user PVE

```
1 root@pve:~# pveum user add terraform@pve --password userTerra
```

Shell

Il faut lui ajouter les bons droits, listés dans le rôle « TerraRole » créé à cet effet (et modifié via l'interface Proxmox, plus simple d'utilisation qu'en CLI)

```
root@pve:~# pveum role add TerraRole -privs "Datastore.AllocateSpace
Datastore.Audit Pool.Allocate Sys.Audit Sys.Console Sys.Modify VM.Allocate
1 VM.Audit VM.Clone VM.Config.CDROM VM.Config.Cloudinit VM.Config.CPU
VM.Config.Disk VM.Config.HWType VM.Config.Memory VM.Config.Network
VM.Config.Options VM.Migrate VM.PowerMgmt SDN.Use"
```

Shell

Roles	Yes	PVEUser	SDN.Audit SDN.Use
Realms	Yes	PVESysAdmin	Sys.Audit Sys.Console Sys.Syslog
HA	Yes	PVETemplateUser	VM.Audit VM.Clone
SDN	Yes	PVEUserAdmin	Group.Allocate Realm.AllocateUser User.Modify
Zones	Yes	PVEVMAdmin	VM.Allocate VM.Audit VM.Backup VM.Clone VM.Config.CDROM VM.Config.CPU VM.Config.Cloudinit VM.Config.Disk VM.Config.HWType VM.Config.Memory VM.Config.Network VM.Config.Options VM.Migrate VM.PowerMgmt VM.Replic VM.Snapshot VM.Snapshot.Rollback
VNets	Yes	PVEVMUser	VM.Audit VM.Backup VM.Config.CDROM VM.Config.Cloudinit VM.Console VM.GuestAgent.Audit VM.GuestAgent.FileRead VM.GuestAgent.FileSystemMgmt VM.GuestAgent.FileWrite VM.GuestAgent.Unrestricted VM.Migrate VM.PowerMgmt VM.Replic
Options	No	TerraRole	Datastore.AllocateSpace Datastore.Audit Pool.Allocate SDN.Use Sys.Audit Sys.Console Sys.Modify VM.Allocate VM.Audit VM.Clone VM.Config.CDROM VM.Config.CPU VM.Config.Cloudinit VM.Config.Disk VM.Config.HWType VM.Config.Memory VM.Config.Network VM.Config.Options VM.Migrate VM.PowerMgmt
IPAM			

Donc, lui ajouter le rôle, et générer le token associé à cet user:

```
root@pve:~# pveum acl modify / -user terraform@pve -role TerraRole
root@pve:~# pveum user token add terraform@pve tf-token --privsep 0
```

key	value
full-tokenid	terraform@pve!tf-token
info	{"privsep":"0"}
value	2ef5d121-2f70-48c7-a17b-78e7dc22d76e

```
root@pve:~# █
```

1.3. Génération des fichiers de configuration Terraform

Les détails de la configuration sont stockés dans le fichier « main.tf »:

```
1  user@debian:~/tform$ cat main.tf
2  provider "proxmox" {
3      pm_api_url      = "https://192.168.122.220:8006/api2/json"
4      pm_api_token_id  = "terraform@pve!tf-token"
5      pm_api_token_secret = "2ef5d121-2f70-48c7-a17b-78e7dc22d76e"
6      pm_tls_insecure   = true
7      pm_minimum_permission_check = false
8  }
9
10 terraform {
11     required_providers {
12         proxmox = {
13             source = "telmate/proxmox"
14             version = "3.0.2-rc07"
15         }
16     }
17 }
18
19 resource "proxmox_vm_qemu" "debian_clone" {
20     name          = "debian-TerraCloned"
21     target_node   = "pve"
22
23     # os_type      = "cloud-init"
24     clone          = "debian"
25     full_clone     = true
26     agent          = 1
27     start_at_node_boot = true
28
29     vmid          = 120
30     memory        = 2048
31 }
```

```
32  cpu {
33    cores      = 2
34    sockets    = 1
35    type       = "host"
36  }
37
38  disk {
39    slot       = "scsi0"
40    size       = "40G"
41    type       = "disk"
42    storage    = "local-lvm"
43  }
44
45  network {
46    id         = 0
47    model      = "virtio"
48    bridge     = "vmbri1"
49  }
50
51 }
```

Les caractéristiques de la VM sont passés en paramètre suivant les consignes. Le seul provider nécessaire à cette étape est Telmate.

1.4. Terraform init et plan

```
user@debian:~/tform$ terraform init -upgrade
Initializing the backend...
```

Initializing provider plugins...

```
- Finding telmate/proxmox versions matching "3.0.2-rc07"...
- Installing telmate/proxmox v3.0.2-rc07...
- Installed telmate/proxmox v3.0.2-rc07 (self-signed, key ID A9EBBE091B35AFCE)
Partner and community providers are signed by their developers.
If you'd like to know more about provider signing, you can read about it here:
https://developer.hashicorp.com/terraform/cli/plugins/signing
```

Terraform has made some changes to the provider dependency selections recorded in the `.terraform.lock.hcl` file. Review those changes and commit them to your version control system if they represent changes you intended to make.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running `"terraform plan"` to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
user@debian:~/tform$
```

```
1 user@debian:~/tform$ terraform plan
2
3 Terraform used the selected providers to generate the following execution plan.
4 Resource actions are indicated with the following symbols:
5   + create
6
7 Terraform will perform the following actions:
8
9   # proxmox_vm_qemu.debian_clone will be created
10  + resource "proxmox_vm_qemu" "debian_clone" {
11    + additional_wait      = 5
12    + agent                = 1
13    + agent_timeout       = 90
14    + automatic_reboot    = true
15    + automatic_reboot_severity = "error"
16    + balloon             = 0
17    + bios                 = "seabios"
18    + boot                 = (known after apply)
19    + bootdisk            = (known after apply)
20    + ciupgrade           = false
21    + clone                = "debian"
22    + clone_wait           = 10
23    + current_node        = (known after apply)
24    + default_ipv4_address = (known after apply)
25    + default_ipv6_address = (known after apply)
26    + define_connection_info = true
27    + description         = "Managed by Terraform."
28    + force_create        = false
29    + full_clone          = true
30    + hotplug              = "network,disk,usb"
31    + id                  = (known after apply)
32    + kvm                  = true
33    + linked_vmid         = (known after apply)
34    + memory              = 2048
35    + name                 = "debian-TerraCloned"
36    + protection          = false
37    + reboot_required     = (known after apply)
38    + scsihw               = "lsi"
39    + skip_ipv4            = false
40    + skip_ipv6            = false
41    + ssh_host             = (known after apply)
42    + ssh_port             = (known after apply)
43    + start_at_node_boot  = true
44    + tablet               = true
45    + target_node          = "pve"
46    + unused_disk         = (known after apply)
```

```
47     + vm_state           = "running"
48     + vmid                = 120
49
50     + cpu {
51         + cores    = 2
52         + limit    = 0
53         + numa     = false
54         + sockets  = 1
55         + type     = "host"
56         + units    = 0
57         + vcores   = 0
58     }
59
60     + disk {
61         + backup      = true
62         + id          = (known after apply)
63         + iops_r_burst = 0
64         + iops_r_burst_length = 0
65         + iops_r_concurrent = 0
66         + iops_wr_burst  = 0
67         + iops_wr_burst_length = 0
68         + iops_wr_concurrent = 0
69         + linked_disk_id = (known after apply)
70         + mbps_r_burst   = 0
71         + mbps_r_concurrent = 0
72         + mbps_wr_burst  = 0
73         + mbps_wr_concurrent = 0
74         + passthrough   = false
75         + size           = "40G"
76         + slot           = "scsi0"
77         + storage        = "local-lvm"
78         + type           = "disk"
79     }
80
81     + network {
82         + bridge    = "vmbri1"
83         + firewall  = false
84         + id        = 0
85         + link_down = false
86         + macaddr   = (known after apply)
87         + model     = "virtio"
88     }
89
90     + smbios (known after apply)
91 }
92
```

```
93 Plan: 1 to add, 0 to change, 0 to destroy.
94
95 _____
96
97 Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to
98 take exactly these actions if you run "terraform apply" now.
```

1.5. Terraform apply

```
user@debian:~/tform$ terraform apply
```

```
Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
+ create
```

```
Terraform will perform the following actions:
```

```
# proxmox_vm_qemu.debian_clone will be created
+ resource "proxmox_vm_qemu" "debian_clone" {
+   additional_wait      = 5
+   agent                = 1
+   agent_timeout        = 90
+   automatic_reboot     = true
+   automatic_reboot_severity = "error"
+   balloon              = 0
+   bios                 = "seabios"
+   boot                 = (known after apply)
+   bootdisk             = (known after apply)
+   ciupgrade            = false
+   clone                = "debian"
+   clone_wait           = 10
+   current_node         = (known after apply)
+   default_ipv4_address = (known after apply)
+   default_ipv6_address = (known after apply)
+   define_connection_info = true
+   description          = "Managed by Terraform."
+   force_create         = false
+   full_clone           = true
+   hotplug              = "network,disk,usb"
+   id                   = (known after apply)
+   kvm                  = true
+   linked_vmid          = (known after apply)
+   memory               = 2048
+   name                 = "debian-TerraCloned"
+   protection           = false
+   reboot_required      = (known after apply)
+   scsihw               = "lsi"
+   skip_ipv4            = false
+   skip_ipv6            = false
+   ssh_host             = (known after apply)
+   ssh_port             = (known after apply)
+   start_at_node_boot   = true
+   tablet               = true
+   target_node          = "pve"
+   unused_disk          = (known after apply)
+   vm_state             = "running"
+   vmid                 = 120

+   cpu {
```



```
+ cores      = 2
+ limit      = 0
+ numa       = false
+ sockets    = 1
+ type       = "host"
+ units      = 0
+ vcores     = 0
}

+ disk {
  + backup            = true
  + id                = (known after apply)
  + iops_r_burst      = 0
  + iops_r_burst_length = 0
  + iops_r_concurrent  = 0
  + iops_wr_burst      = 0
  + iops_wr_burst_length = 0
  + iops_wr_concurrent = 0
  + linked_disk_id     = (known after apply)
  + mbps_r_burst       = 0
  + mbps_r_concurrent  = 0
  + mbps_wr_burst       = 0
  + mbps_wr_concurrent = 0
  + passthrough        = false
  + size               = "40G"
  + slot               = "scsi0"
  + storage             = "local-lvm"
  + type               = "disk"
}

+ network {
  + bridge    = "vmbr1"
  + firewall  = false
  + id        = 0
  + link_down = false
  + macaddr   = (known after apply)
  + model     = "virtio"
}

+ smbios (known after apply)
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
proxmox_vm_qemu.debian_clone: Creating...
proxmox_vm_qemu.debian_clone: Still creating... [00m10s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m20s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m30s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m40s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m50s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m00s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m10s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m20s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m30s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m40s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m50s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [02m00s elapsed]
proxmox_vm_qemu.debian_clone: Creation complete after 2m2s [id=pve/qemu/120]
```

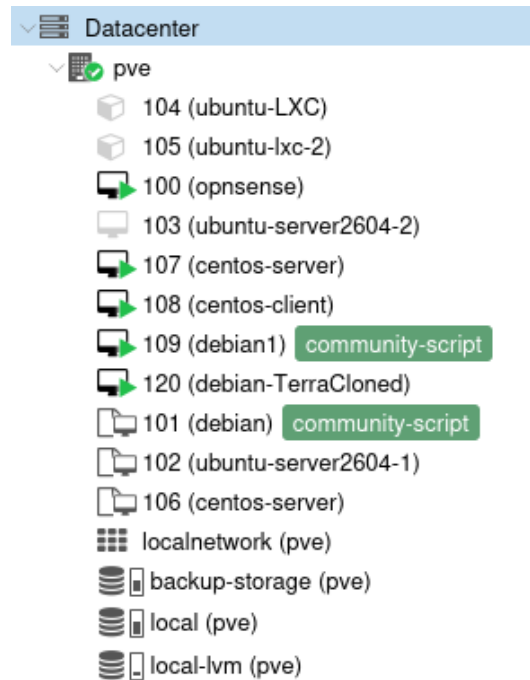
Warning: Qemu Guest Agent is enabled but no IPv6 address is found

```
with proxmox_vm_qemu.debian_clone,
on main.tf line 18, in resource "proxmox_vm_qemu" "debian_clone":
18: resource "proxmox_vm_qemu" "debian_clone" {
```

Qemu Guest Agent is enabled in your configuration but no IPv6 address was found before the time ran out, increasing 'agent_timeout' could resolve this issue.
To suppress this warning set 'skip_ipv6' to true.

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Et la VM créée (id 120):



1.6. Terraform destroy

```
user@debian:~/tform$ terraform destroy
proxmox_vm_qemu.debian_clone: Refreshing state... [id=pve/qemu/120]
```

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:

- destroy

Terraform will perform the following actions:

```
# proxmox_vm_qemu.debian_clone will be destroyed
- resource "proxmox_vm_qemu" "debian_clone" {
  - additional_wait      = 5 -> null
  - agent               = 1 -> null
  - agent_timeout       = 90 -> null
  - automatic_reboot    = true -> null
  - automatic_reboot_severity = "error" -> null
  - balloon             = 0 -> null
  - bios                = "seabios" -> null
  - boot                = "order=scsi0" -> null
  - ciupgrade           = false -> null
  - clone               = "debian" -> null
  - clone_wait          = 10 -> null
  - cores               = 0 -> null
  - current_node        = "pve" -> null
  - default_ipv4_address = "10.0.0.68" -> null
  - define_connection_info = true -> null
  - description         = "Managed by Terraform." -> null
  - force_create        = false -> null
  - full_clone          = true -> null
  - hotplug             = "network,disk,usb" -> null
  - id                  = "pve/qemu/120" -> null
  - kvm                 = true -> null
  - linked_vmid         = 0 -> null
  - memory              = 2048 -> null
  - name                = "debian-TerraCloned" -> null
  - numa               = false -> null
  - protection          = false -> null
  - qemu_os             = "l26" -> null
  - reboot_required     = false -> null
  - scsihw              = "lsi" -> null
  - skip_ipv4           = false -> null
  - skip_ipv6           = false -> null
  - sockets             = 0 -> null
  - ssh_host            = "10.0.0.68" -> null
  - ssh_port            = "22" -> null
  - start_at_node_boot  = true -> null
  - tablet              = true -> null
  - tags                = null
  - target_node         = "pve" -> null
```

```
- unused_disk      = [] -> null
- vcpus            = 0 -> null
- vm_state         = "running" -> null
- vmid            = 120 -> null
# (10 unchanged attributes hidden)

- cpu {
  - cores      = 2 -> null
  - limit      = 0 -> null
  - numa       = false -> null
  - sockets    = 1 -> null
  - type       = "host" -> null
  - units      = 0 -> null
  - vcores     = 0 -> null
# (1 unchanged attribute hidden)
}

- disk {
  - backup      = true -> null
  - discard     = false -> null
  - emulatsd    = false -> null
  - format      = "raw" -> null
  - id          = 1 -> null
  - iops_r_burst = 0 -> null
  - iops_r_burst_length = 0 -> null
  - iops_r_concurrent = 0 -> null
  - iops_wr_burst = 0 -> null
  - iops_wr_burst_length = 0 -> null
  - iops_wr_concurrent = 0 -> null
  - iothread    = false -> null
  - linked_disk_id = -1 -> null
  - mbps_r_burst = 0 -> null
  - mbps_r_concurrent = 0 -> null
  - mbps_wr_burst = 0 -> null
  - mbps_wr_concurrent = 0 -> null
  - passthrough = false -> null
  - readonly    = false -> null
  - replicate   = false -> null
  - size        = "40G" -> null
  - slot        = "scsi0" -> null
  - storage     = "local-lvm" -> null
  - type        = "disk" -> null
# (6 unchanged attributes hidden)
}

- network {
  - bridge      = "vmbr1" -> null
  - firewall    = false -> null

  - id          = 0 -> null
  - link_down   = false -> null
  - macaddr     = "bc:24:11:a4:aa:43" -> null
  - model       = "virtio" -> null
  - mtu         = 0 -> null
  - queues      = 0 -> null
  - rate        = 0 -> null
  - tag         = 0 -> null
}

- smbios {
  - uuid        = "b455933a-67c2-4bb3-943e-7ff6f2d19300" -> null
# (6 unchanged attributes hidden)
}

- startup_shutdown {
  - order       = -1 -> null
  - shutdown_timeout = -1 -> null
  - startup_delay = -1 -> null
}
}
```

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

proxmox_vm_qemu.debian_clone: Destroying... [id=pve/qemu/120]
proxmox_vm_qemu.debian_clone: Destruction complete after 4s

Destroy complete! Resources: 1 destroyed.
user@debian:~/terraform

Cela fonctionne parfaitement.

2. Déploiement avec adresse MAC fixée

2.1. DNSMasq to KEA & Unbound

Arrêt du DNSMasq:

The screenshot shows the OPNsense web interface. On the left is a sidebar menu with the 'Services' section expanded, showing 'DNS & DHCP Dnsmasq' as the selected option. The main content area is titled 'Services: DNS & DHCP Dnsmasq' and has several tabs: 'Général', 'Domaines', 'Hôtes', 'Plages DHCP', 'Options DHCP', 'DHCP boot', and 'Balises DHCP'. The 'Général' tab is active. It shows a 'mode avancé' toggle. Under the 'Défaut' section, there is an 'Activer' checkbox (unchecked) and an 'Interface' dropdown menu set to 'LAN'. Below this, the 'DNS' section includes a 'Port d'écoute' field set to '53053' and several checkboxes for DNSSEC, Pas de recherche d'hôtes, Expand hosts, and Transfert de requête DNS. The 'Transfert de requête DNS' section has four checkboxes: Interrogation séquentielle des serveurs DNS, Exiger un domaine, Ne pas transmettre aux serveurs DNS définis par le système, and Ne pas faire suivre les recherches locales inversées. All checkboxes in this section are unchecked.

Activation de KEA Agent:

The screenshot shows the 'Services: Kea DHCP: Agent de contrôle' configuration page. It has a 'mode avancé' toggle and an 'aide complète' link. The 'Activé' checkbox is checked. Below it, the 'Adresse de liaison' field is set to '127.0.0.1' and the 'Port de liaison' field is set to '8000'. At the bottom, there is an 'Appliquer' button.

Activation de KEA DHCP:

The screenshot shows the 'Services: Kea DHCP: Kea DHCPv4' configuration page. It has tabs for 'Paramètres', 'Sous-réseaux', 'Réservations', 'Options', and 'Paires de l'AP'. The 'Paramètres' tab is active. It shows a 'mode avancé' toggle and an 'aide complète' link. Under the 'Service' section, the 'Activé' checkbox is checked. Below this, the 'Paramètres généraux' section includes an 'Interfaces' dropdown menu set to 'LAN', a 'Durée de vie valide' field set to '4000', a 'Règles de pare-feu' checkbox (checked), and a 'Type de prise' dropdown menu set to 'raw'. There are also expandable sections for 'Lease Expiration' and 'Haute disponibilité'. At the bottom, there is an 'Appliquer' button.

Définition du sous-réseau visé:

Services: Kea DHCP: Kea DHCPv4

Paramètres

Sous-réseaux

Réervations

Options

Pairs de l'AP

Recherche

50

☐	Sous-réseau	Description	Piscines	Actions
☐	10.0.0.0/24	LAN	10.0.0.41 - 10.0.0.245	

<

<

1

>

>

Affichage des entrées 1 à 1 sur 1

Appliquer

Activation du service Unbound:

Services: Unbound DNS: Général

mode avancé

aide complète

1 Activer Unbound

☒

2 Port d'écoute

53

3 Interfaces réseau

Tout (recommandé)

Tout effacer Sélectionner tout

4 Activer le support DNSSEC

☐

5 Activer le support DNS64

☐

6 Préfixe DNS64

64:ff9b::/96

7 Activer le mode AAAA uniquement

☐

8 Enregistrer les baux DHCP4 ICS

☐

9 Contournement de domaine DHCP

10 Enregistrer les correspondances statiques DHCP

☐

11 Ne pas enregistrer les adresses IPv6 Link-Local

☐

12 Ne pas enregistrer les enregistrements du système A/AAAA

☐

13 Prise en charge des commentaires TXT

☐

14 Vider le cache DNS lors du rechargement

☐

15 Forcer SafeSearch

☐

16 Type de zone locale

transparent

Appliquer

Allocation dynamique d'IP fonctionnelle avec KEA (machine redémarrée):

ubuntu-server2604-2 (Uptime: 00:01:18)

Status

running

HA State

none

Node

pve

CPU usage

2.75% of 1 CPU

Memory usage

14.73% (603.36 MiB of 4.00 GiB)

Host memory usage

1.19 GiB

Bootdisk size

32.00 GiB

IPs

10.0.0.41
fe80::be24:11ff:fe25:a21b

More

Définition dynamique du DNS et Gateway:

Services: Kea DHCP: Kea DHCPv4

Paramètres Sous-réseaux Réservations Options Pairs de l'AP

Recherche

☐	Description	Match Code	Match Data	Set Code	Set Data	Actions
☐	dns			name server [5]	10.0.0.1	
☐	gateway			router [3]	10.0.0.1	

« < 1 > »

Affichage des entrées 1 à 2 sur 2

2.2. Génération d'une adresse MAC statique

Ajout d'Ivoronin comme provider et génération de la ressource MAC address:

```
1 # dans main.tf
2
3 terraform {
4   required_providers {
5     proxmox = {
6       source = "telmate/proxmox"
7       version = "3.0.2-rc04" #3.0.2-rc07
8     }
9   }
10   macaddress = {
11     source = "ivoronin/macaddress"
12     version = "0.3.0"
13   }
14 }
15 [...]
16
17 resource "macaddress" "vmmac" {
18 }
```

Puis, utilisation de cette adresse MAC dans les spécifications réseau du clone:

```
1 # dans main.tf
2
3 network {
4   id      = 0
5   model   = "virtio"
6   bridge  = "vmbri1"
7   macaddr = upper(macaddress.vmmac.address)
8 }
```

En uppercase suivant les exigences de proxmox.

2.3. Attribution d'une IP statique à l'adresse MAC de la VM clonée

Pour cela il faut créer un group + user pour récupérer un token d'accès API à Opnsense:

Système: Accès: Groupes

Recherche

50

☐	Nom du groupe	Privilèges	Membres	Description	Réseaux source	Actions
☐	admins	All pages	root	System Administrators		
☐	tf-group	Services: DHCP: Kea Ctrl Age...	terraform			

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Affichage des entrées 1 à 2 sur 2

avec les privilèges de contrôle de KEA. Puis, attribuer ce rôle à un user pour le quel on pourra récupérer une clé API:

Système: Accès: Utilisateurs

Utilisateurs

Clés Api

Recherche

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☐	Nom d'utilisateur	Membre du groupe	Nom complet	Actions
☐	 root	admins	System Administrator	   
☐	 terraform	tf-group		   

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Affichage des entrées 1 à 2 sur 2

Système: Accès: Utilisateurs

Utilisateurs

Clés Api

Recherche

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☐	Nom d'utilisateur	Clé Api	Actions
☐	terraform	AncmTmUYXl5csabprgylFjDahbfRohVt7jYNm+UbKnPwAapQXaHXV7K3/...	

«

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Affichage des entrées 1 à 1 sur 1

Cette clé peut-être ajoutée dans le provider Browningluke qui s'interface avec Opnsense:

```
1 # dans main.tf
2
3 provider "opnsense" {
4   uri = "https://192.168.122.197"
5   api_key = "AncmTmUYXl5csabprgylFjDahbfRohVt7jYNm+UbKnPwAapQXaHXV7K3/UeYLUkcFY>
6   api_secret = "JU4/5kxzlFm5tALdiscA5Bzt1s6F+TZh6GoY2JLf0ITYUVadoVs4CjEq8lGH2kV>
7   allow_insecure = true
8 }
```

avec allow_insecure pour contourner l'autovalidation du certificat.

puis:

```
1 # dans main.tf
2
3 resource "opnsense_kea_dhcpv4_reservation" "debian_clone" {
4     subnet_id    = "4f275fc3-7f2d-4bdb-b8a7-eb794075aa7f"
5     ip_address   = "10.0.0.140"
6     mac_address  = lower(macaddress.vmmac.address)
7     hostname     = "tf_cloned_debian"
8     description  = "Static IP for tf_cloned_debian"
9 }
```

Shell

Le subnet ID correspond à l'UUID du sous réseau créé précédemment de plage 10.0.0.41 - 10.0.0.245. L'adresse IP fixe attribuée à l'adresse MAC est 10.0.0.140 et l'adresse mac est passée en lowercase (contrairement à OpnSense qui utilise l'Uppercase...)

2.4. Terraform plan

```
Destroy complete! Resources: 3 destroyed.
user@debian:~/tfom$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# macaddress.vmmac will be created
+ resource "macaddress" "vmmac" {
  + address = (known after apply)
  + id      = (known after apply)
}

# opnsense_kea_dhcpv4_reservation.debian_clone will be created
+ resource "opnsense_kea_dhcpv4_reservation" "debian_clone" {
  + description = "Static IP for tf_cloned_debian"
  + hostname    = "tf_cloned_debian"
  + id          = (known after apply)
  + ip_address  = "10.0.0.140"
  + mac_address = (known after apply)
  + subnet_id   = "4f275fc3-7f2d-4bdb-b8a7-eb794075aa7f"
}

# proxmox_vm_qemu.debian_clone will be created
+ resource "proxmox_vm_qemu" "debian_clone" {
  + additional_wait = 5
  + agent           = 1
  + agent_timeout   = 90
  + automatic_reboot = true
  + balloon         = 0
  + bios            = "seabios"
  + boot            = (known after apply)
  + bootdisk        = (known after apply)
  + ciupgrade       = false
  + clone           = "debian"
  + clone_wait      = 10
  + current_node     = (known after apply)
  + default_ipv4_address = (known after apply)
  + default_ipv6_address = (known after apply)
  + define_connection_info = true
  + description      = "Managed by Terraform."
  + force_create     = false
  + full_clone       = true
  + hotplug          = "network,disk,usb"
  + id               = (known after apply)
  + kvm              = true
}
```



```
+ linked_vmid      = (known after apply)
+ memory          = 2048
+ name            = "debian-TerraCloned"
+ onboot          = true
+ protection       = false
+ reboot_required = (known after apply)
+ scsihw          = "lsi"
+ skip_ipv4        = false
+ skip_ipv6        = false
+ ssh_host        = (known after apply)
+ ssh_port        = (known after apply)
+ tablet          = true
+ target_node     = "pve"
+ unused_disk     = (known after apply)
+ vm_state        = "running"
+ vmid            = 120

+ cpu {
+   cores    = 2
+   limit    = 0
+   numa     = false
+   sockets  = 1
+   type     = "host"
+   units    = 0
+   vcores   = 0
+ }

+ disk {
+   backup      = true
+   id          = (known after apply)
+   iops_r_burst      = 0
+   iops_r_burst_length = 0
+   iops_r_concurrent = 0
+   iops_wr_burst      = 0
+   iops_wr_burst_length = 0
+   iops_wr_concurrent = 0
+   linked_disk_id    = (known after apply)
+   mbps_r_burst      = 0
+   mbps_r_concurrent = 0
+   mbps_wr_burst      = 0
+   mbps_wr_concurrent = 0
+   passthrough       = false
+   size               = "40G"
+   slot               = "scsi0"
+   storage            = "local-lvm"
+   type               = "disk"
+ }

+ network {

+   bridge    = "vmbr1"
+   firewall  = false
+   id        = 0
+   link_down = false
+   macaddr   = (known after apply)
+   model     = "virtio"
+ }

+ smbios (known after apply)
}
```

Plan: 3 to add, 0 to change, 0 to destroy.

Note: You didn't use the `-out` option to save this plan, so Terraform can't guarantee to take exactly these actions if you run `"terraform apply"` now.
`user@debian:~/tforms$`

2.5. Terraform apply

```
user@debian:~/terraform$ terraform apply
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# macaddress.vmmac will be created
+ resource "macaddress" "vmmac" {
  + address = (known after apply)
  + id      = (known after apply)
}

# opnsense_kea_dhcpv4_reservation.debian_clone will be created
+ resource "opnsense_kea_dhcpv4_reservation" "debian_clone" {
  + description = "Static IP for tf_cloned_debian"
  + hostname    = "tf_cloned_debian"
  + id          = (known after apply)
  + ip_address  = "10.0.0.140"
  + mac_address = (known after apply)
  + subnet_id   = "4f275fc3-7f2d-4bdb-b8a7-eb794075aa7f"
}

# proxmox_vm_qemu.debian_clone will be created
+ resource "proxmox_vm_qemu" "debian_clone" {
  + additional_wait = 5
  + agent           = 1
  + agent_timeout   = 90
  + automatic_reboot = true
  + balloon         = 0
  + bios            = "seabios"
  + boot            = (known after apply)
  + bootdisk        = (known after apply)
  + ciupgrade       = false
  + clone           = "debian"
  + clone_wait      = 10
  + current_node     = (known after apply)
  + default_ipv4_address = (known after apply)
  + default_ipv6_address = (known after apply)
  + define_connection_info = true
  + description      = "Managed by Terraform."
  + force_create     = false
  + full_clone       = true
  + hotplug          = "network,disk,usb"
  + id               = (known after apply)
  + kvm              = true
  + linked_vmid      = (known after apply)
  + memory           = 2048
}
```

```
+ name           = "debian-TerraCloned"
+ onboot         = true
+ protection     = false
+ reboot_required = (known after apply)
+ scsihw         = "lsi"
+ skip_ipv4      = false
+ skip_ipv6      = false
+ ssh_host       = (known after apply)
+ ssh_port       = (known after apply)
+ tablet         = true
+ target_node    = "pve"
+ unused_disk    = (known after apply)
+ vm_state       = "running"
+ vmid           = 120

+ cpu {
  + cores    = 2
  + limit    = 0
  + numa     = false
  + sockets  = 1
  + type     = "host"
  + units    = 0
  + vcores   = 0
}

+ disk {
  + backup           = true
  + id               = (known after apply)
  + iops_r_burst     = 0
  + iops_r_burst_length = 0
  + iops_r_concurrent = 0
  + iops_wr_burst     = 0
  + iops_wr_burst_length = 0
  + iops_wr_concurrent = 0
  + linked_disk_id   = (known after apply)
  + mbsps_r_burst     = 0
  + mbsps_r_concurrent = 0
  + mbsps_wr_burst     = 0
  + mbsps_wr_concurrent = 0
  + passthrough       = false
  + size              = "40G"
  + slot              = "scsi0"
  + storage           = "local-lvm"
  + type              = "disk"
}

+ network {
  + bridge    = "vmbr1"
  + firewall  = false
  + id        = 0
  + link_down = false

  + macaddr = (known after apply)
  + model   = "virtio"
}

+ smbios (known after apply)
}
```

Plan: 3 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
macaddress.vmmac: Creating...
macaddress.vmmac: Creation complete after 0s [id=16:43:7e:85:3b:8c]
opnsense_kea_dhcpv4_reservation.debian_clone: Creating...
proxmox_vm_qemu.debian_clone: Creating...
opnsense_kea_dhcpv4_reservation.debian_clone: Creation complete after 0s [id=9d6ad7c3-0e26-4b9f-8e56-ad5271bc2399]
proxmox_vm_qemu.debian_clone: Still creating... [00m10s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m20s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m30s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m40s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m50s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m00s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m10s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m20s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m30s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m40s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [01m50s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [02m00s elapsed]
proxmox_vm_qemu.debian_clone: Creation complete after 2m4s [id=pve/qemu/120]
```

Warning: Qemu Guest Agent is enabled but no IPv6 address is found

```
with proxmox_vm_qemu.debian_clone,
on main.tf line 37, in resource "proxmox_vm_qemu" "debian_clone":
37: resource "proxmox_vm_qemu" "debian_clone" {
```

Qemu Guest Agent is enabled in your configuration but no IPv6 address was found before the time ran out, increasing 'agent_timeout' could resolve this issue. To suppress this warning set 'skip_ipv6' to true.

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
user@debian:~/tform\$

Il est possible d'ajouter le tag skip_ipv6 pour supprimer le warning qui pop à chaque Apply.
Sur proxmox:

Virtual Machine 120 (debian-TerraCloned) on node 'pve'No Tags

Summary

ConsoleHardwareCloud-InitOptionsTask HistoryMonitorBackupReplicationSnapshotsFirewallPermissions

debian-TerraCloned (Uptime: 00:02:54)

Statusrunning

HA Statenone

Nodepve

CPU usage0.66% of 2 CPUs

Memory usage21.85% (447.57 MiB of 2.00 GiB)

Host memory usage447.57 MiB

Bootdisk size40.00 GiB

IPs10.0.0.140
fe80::1443:7eff:fe85:3b8c

NotesManaged by Ter

Network Information

Name	MAC address	IP address
lo	00:00:00:00:00:00	127.0.0.1/8 ::1/128
ens18	16:43:7e:85:3b:8c	10.0.0.140/24 fe80::1443:7eff:fe85:3b8c/64

On constate bien que l'adresse MAC générée (id=16:43:7e:85:3b:8c) a été correctement, attribuée, ainsi que l'adresse IP, confirmé par OpnSense:

Services: Kea DHCP: Kea DHCPv4

ParamètresSous-réseauxRéservationsOptionsPairs de l'AP

Recherche

50

	Sous-réseau	Adresse IP	Adresse MAC	ID Client	Nom d'hôte	Description	Actions
	10.0.0.0/24	10.0.0.140	16:43:7e:85:3b:8c		tf_cloned_debian	Static IP for tf_cloned_...	

Affichage des entrées 1 à 1 sur 1

Appliquer

2.6. Terraform destroy

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

```
user@debian:~/tform$ terraform destroy
macaddress.vmmac: Refreshing state... [id=16:43:7e:85:3b:8c]
opnsense_kea_dhcpv4_reservation.debian_clone: Refreshing state... [id=9d6ad7c3-0e26-4b9f-8e56-ad5271bc2399]
proxmox_vm_qemu.debian_clone: Refreshing state... [id=pve/qemu/120]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- destroy

Terraform will perform the following actions:

```
# macaddress.vmmac will be destroyed
- resource "macaddress" "vmmac" {
  - address = "16:43:7e:85:3b:8c" -> null
  - id      = "16:43:7e:85:3b:8c" -> null
}

# opnsense_kea_dhcpv4_reservation.debian_clone will be destroyed
- resource "opnsense_kea_dhcpv4_reservation" "debian_clone" {
  - description = "Static IP for tf_cloned_debian" -> null
  - hostname   = "tf_cloned_debian" -> null
  - id         = "9d6ad7c3-0e26-4b9f-8e56-ad5271bc2399" -> null
  - ip_address = "10.0.0.140" -> null
  - mac_address = "16:43:7e:85:3b:8c" -> null
  - subnet_id   = "4f275fc3-7f2d-4bdb-b8a7-eb794075aa7f" -> null
}

# proxmox_vm_qemu.debian_clone will be destroyed
```

Plan: 0 to add, 0 to change, 3 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

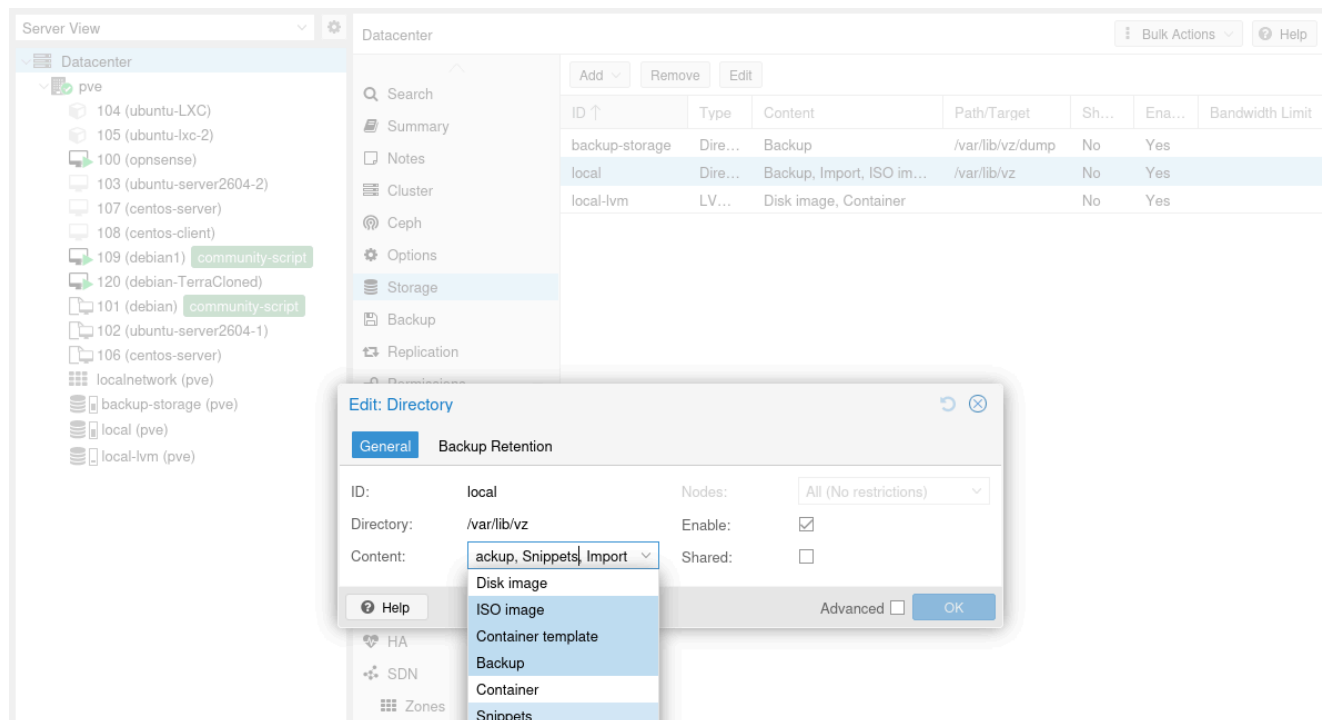
```
opnsense_kea_dhcpv4_reservation.debian_clone: Destroying... [id=9d6ad7c3-0e26-4b9f-8e56-ad5271bc2399]
proxmox_vm_qemu.debian_clone: Destroying... [id=pve/qemu/120]
opnsense_kea_dhcpv4_reservation.debian_clone: Destruction complete after 1s
proxmox_vm_qemu.debian_clone: Destruction complete after 4s
macaddress.vmmac: Destroying... [id=16:43:7e:85:3b:8c]
macaddress.vmmac: Destruction complete after 0s
```

Destroy complete! Resources: 3 destroyed.

```
user@debian:~/tform$
```

3. Implémentation de Cloud-init

3.1. Activation snippets dans local



3.2. Création d'un yaml pour Cloud-init

```
1 #cloud-config
2 hostname: sti-web01
3
4 users:
5   - default
6   - name: ${ci_user}
7     groups: [sudo]
8     shell: /bin/bash
9     sudo: ["ALL=(ALL) NOPASSWD:ALL"]
10    ssh_authorized_keys:
11      - ${ssh_public_key}
12
13 package_update: true
14 package_upgrade: true
15 packages:
16   - vim
17   - curl
18   - nginx
19
20 runcmd:
21   - systemctl enable qemu-guest-agent
22   - systemctl start qemu-guest-agent
```

Les package installés sont vim, curl et nginx.

3.3. Gestion de l'envoi du yml dans local:snippets

```
1  # dans main.tf
2
3  resource "terraform_data" "upload_snippet" {
4    triggers_replace = {
5      snippet_hash = filemd5("user-data.yml")
6    }
7
8    connection {
9      type      = "ssh"
10     user       = "root"
11     private_key = file(pathexpand("~/ssh/id_ed25519"))
12     host       = "192.168.122.220"
13   }
14
15   provisioner "remote-exec" {
16     inline = [
17       "mkdir -p /var/lib/vz/snippets/"
18     ]
19   }
20
21   provisioner "file" {
22     source      = "user-data.yml"
23     destination = "/var/lib/vz/snippets/user-data.yml"
24   }
25 }
```

Il faut donc:

- Vérifier (avec le hash) si le fichier de config est différent de celui présent sur local:snippets
- Ouvre une session SSH root
- Crée le dossier /var/lib/vz/snippets/ dans le disque local (-p pour ne pas renvoyer d'erreur s'il existe déjà)
- Effectue la copie via scp

3.4. Dernières modifications du main.tf

```
1  # dans main.tf
2
3  resource "proxmox_vm_qemu" "debian_clone" {
4    name          = "debian-TerraCloned"
5    target_node   = "pve"
6
7    depends_on    = [terraform_data.upload_snippet] #ajout de la dépendance
8
9    os_type       = "cloud-init" #modif ici
10    ciuser        = var.ci_user #les variables ciuser et ssh_public_key_path sont stockées
11                  #dans un var.tf associé, qui servira à stocker l'ensemble des variables pour + de
12                  #propreté
13    sshkeys       = trimspace(file(pathexpand(var.ssh_public_key_path)))
```

```
12  clone      = "debian"
13  full_clone = true
14  agent      = 1
15  start_at_node_boot = true
16  # onboot = true
17  ipconfig0 = "ip=dhcp"
18  cicustom = "user=local:snippets/user-data.yml" #ajout ici
```


3.5. Terraform plan

```
user@débien:~/tform$ terraform plan
```

Terraform used the selected providers to generate the following execution plan. For actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# macaddress.vmmac will be created
+ resource "macaddress" "vmmac" {
  + address = (known after apply)
  + id      = (known after apply)
}

# opnsense_kea_dhcpv4_reservation.debian_clone will be created
+ resource "opnsense_kea_dhcpv4_reservation" "debian_clone" {
  + description = "Static IP for tf_cloned_debian"
  + hostname    = "tf_cloned_debian"
  + id          = (known after apply)
  + ip_address  = "10.0.0.140"
  + mac_address = (known after apply)
  + subnet_id   = "4f275fc3-7f2d-4bdb-b8a7-eb794075aa7f"
}

# proxmox_vm_qemu.debian_clone will be created
+ resource "proxmox_vm_qemu" "debian_clone" {
  + additional_wait = 5
  + agent           = 1
  + agent_timeout   = 90
  + automatic_reboot = true
  + automatic_reboot_severity = "error"
  + balloon         = 0
  + bios            = "seabios"
  + boot            = (known after apply)
  + bootdisk        = (known after apply)
  + cicustom        = "user=local:snippets/user-data.yml"
  + ciupgrade       = false
  + ciuser          = "admin"
  + clone           = "debian"
  + clone_wait      = 10
  + current_node    = (known after apply)
  + default_ipv4_address = (known after apply)
  + default_ipv6_address = (known after apply)
  + define_connection_info = true
  + description     = "Managed by Terraform."
  + force_create    = false
  + full_clone      = true
  + hotplug         = "network,disk,usb"
  + id              = (known after apply)
  + ipconfig0       = "ip=dhcp"
  + kvm             = true
}
```

```
+ linked_vmid      = (known after apply)
+ memory          = 2048
+ name            = "debian-TerraCloned"
+ os_type         = "cloud-init"
+ protection      = false
+ reboot_required = (known after apply)
+ scsihw          = "lsi"
+ skip_ipv4       = false
+ skip_ipv6       = true
+ ssh_host        = (known after apply)
+ ssh_port        = (known after apply)
+ sshkeys         = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDcr42Ct
JQHGM4c0CQPjYyT0W3W74RtL0ydw user@debian"
+ start_at_node_boot = true
+ tablet          = true
+ target_node     = "pve"
+ unused_disk     = (known after apply)
+ vm_state        = "running"
+ vmid            = 120

+ cpu {
+   cores    = 2
+   limit    = 0
+   numa     = false
+   sockets  = 1
+   type     = "host"
+   units    = 0
+   vcores   = 0
+ }

+ disk {
+   backup          = true
+   id              = (known after apply)
+   iops_r_burst    = 0
+   iops_r_burst_length = 0
+   iops_r_concurrent = 0
+   iops_wr_burst   = 0
+   iops_wr_burst_length = 0
+   iops_wr_concurrent = 0
+   linked_disk_id  = (known after apply)
+   mbps_r_burst    = 0
+   mbps_r_concurrent = 0
+   mbps_wr_burst   = 0
+   mbps_wr_concurrent = 0
+   passthrough     = false
+   size            = "40G"
+   slot            = "scsi0"
+   storage         = "local-lvm"

+   + type = "disk"
+ }

+ network {
+   bridge    = "vmbri"
+   firewall  = false
+   id        = 0
+   link_down = false
+   macaddr   = (known after apply)
+   model     = "virtio"
+ }

+ smbios (known after apply)
}

# terraform_data.upload_snippet will be created
+ resource "terraform_data" "upload_snippet" {
+   id = (known after apply)
+   triggers_replace = {
+     snippet_hash = "76630081873a374637e8471c617d469e"
+   }
+ }
```

Plan: 4 to add, 0 to change, 0 to destroy.

Note: You didn't use the `-out` option to save this plan, so Terraform can't guarantee exactly these actions if you run `"terraform apply"` now.

`user@debian:~/tform$` 

3.6. Terraform apply

```
user@debian:~/tform$ terraform apply
```

Terraform used the selected providers to generate the following execution plan. For actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# macaddress.vmmac will be created
+ resource "macaddress" "vmmac" {
  + address = (known after apply)
  + id      = (known after apply)
}

# opnsense_kea_dhcpv4_reservation.debian_clone will be created
+ resource "opnsense_kea_dhcpv4_reservation" "debian_clone" {
  + description = "Static IP for tf_cloned_debian"
  + hostname    = "tf_cloned_debian"
  + id          = (known after apply)
  + ip_address  = "10.0.0.140"
  + mac_address = (known after apply)
  + subnet_id   = "4f275fc3-7f2d-4bdb-b8a7-eb794075aa7f"
}

# proxmox_vm_qemu.debian_clone will be created
+ resource "proxmox_vm_qemu" "debian_clone" {
  + additional_wait      = 5
  + agent                = 1
  + agent_timeout        = 90
  + automatic_reboot     = true
  + automatic_reboot_severity = "error"
  + balloon              = 0
  + bios                 = "seabios"
  + boot                 = (known after apply)
  + bootdisk             = (known after apply)
  + cicustom             = "user=local:snippets/user-data.yml"
  + ciupgrade            = false
  + ciuser               = "admin"
  + clone                = "debian"
  + clone_wait           = 10
  + current_node          = (known after apply)
  + default_ipv4_address = (known after apply)
  + default_ipv6_address = (known after apply)
  + define_connection_info = true
  + description          = "Managed by Terraform."
  + force_create          = false
  + full_clone           = true
  + hotplug              = "network,disk,usb"
  + id                   = (known after apply)
  + ipconfig0            = "ip=dhcp"
```

```
+ kvm = true
+ linked_vmid = (known after apply)
+ memory = 2048
+ name = "debian-TerraCloned"
+ os_type = "cloud-init"
+ protection = false
+ reboot_required = (known after apply)
+ scsihw = "lsi"
+ skip_ipv4 = false
+ skip_ipv6 = true
+ ssh_host = (known after apply)
+ ssh_port = (known after apply)
+ sshkeys = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDcr42Cr
JQHGM4c0CQPjYyT0W3W74RtL0ydw user@debian"
+ start_at_node_boot = true
+ tablet = true
+ target_node = "pve"
+ unused_disk = (known after apply)
+ vm_state = "running"
+ vmid = 120

+ cpu {
  + cores = 2
  + limit = 0
  + numa = false
  + sockets = 1
  + type = "host"
  + units = 0
  + vcores = 0
}

+ disk {
  + backup = true
  + id = (known after apply)
  + iops_r_burst = 0
  + iops_r_burst_length = 0
  + iops_r_concurrent = 0
  + iops_wr_burst = 0
  + iops_wr_burst_length = 0
  + iops_wr_concurrent = 0
  + linked_disk_id = (known after apply)
  + mbps_r_burst = 0
  + mbps_r_concurrent = 0
  + mbps_wr_burst = 0
  + mbps_wr_concurrent = 0
  + passthrough = false
  + size = "40G"
  + slot = "scsi0"
```

```
+ storage          = "local-lvm"
+ type             = "disk"
}

+ network {
  + bridge      = "vmbri1"
  + firewall    = false
  + id          = 0
  + link_down   = false
  + macaddr     = (known after apply)
  + model       = "virtio"
}

+ smbios (known after apply)
}

# terraform_data.upload_snippet will be created
+ resource "terraform_data" "upload_snippet" {
  + id          = (known after apply)
  + triggers_replace = {
    + snippet_hash = "76630081873a374637e8471c617d469e"
  }
}
```

Plan: 4 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
terraform_data.upload_snippet: Creating...
terraform_data.upload_snippet: Provisioning with 'remote-exec'...
terraform_data.upload_snippet (remote-exec): Connecting to remote host via SSH...
terraform_data.upload_snippet (remote-exec): Host: 192.168.122.220
terraform_data.upload_snippet (remote-exec): User: root
terraform_data.upload_snippet (remote-exec): Password: false
terraform_data.upload_snippet (remote-exec): Private key: true
terraform_data.upload_snippet (remote-exec): Certificate: false
terraform_data.upload_snippet (remote-exec): SSH Agent: true
terraform_data.upload_snippet (remote-exec): Checking Host Key: false
terraform_data.upload_snippet (remote-exec): Target Platform: unix
macaddress.vmmac: Creating...
macaddress.vmmac: Creation complete after 0s [id=e6:a6:fb:1c:85:04]
opnsense_kea_dhcpv4_reservation.debian_clone: Creating...
opnsense_kea_dhcpv4_reservation.debian_clone: Creation complete after 1s [id=926c
f-4996-b29e-21d7df1c5490]
```

```
terraform_data.upload_snippet (remote-exec): Connected!
terraform_data.upload_snippet: Provisioning with 'file'...
terraform_data.upload_snippet: Creation complete after 1s [id=61d91797-848e-6914-401291358]
proxmox_vm_qemu.debian_clone: Creating...
proxmox_vm_qemu.debian_clone: Still creating... [00m10s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m20s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m30s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m40s elapsed]
proxmox_vm_qemu.debian_clone: Still creating... [00m50s elapsed]
proxmox_vm_qemu.debian_clone: Creation complete after 55s [id=pve/qemu/120]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.
user@debian:~/tform$
```

vérification:

```
root@localhost:~# curl 127.0.0.1 > output.txt && head -n 5 output.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100 10701  100 10701    0     0 1639k      0 --:--:-- --:--:-- --:--:-- 1741k

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
root@localhost:~#
```

cURL fonctionne parfaitement

```
root@localhost:~# vim output.txt > tqf.txt
Vim: Warning: Output is not to a terminal
```

Vim aussi

```
root@localhost:~# sudo apt update && apt upgrade
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:2 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Hit:3 https://deb.debian.org/debian bookworm InRelease
Get:4 https://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:5 https://deb.debian.org/debian bookworm-backports InRelease [59.4 kB]
Get:6 https://deb.debian.org/debian-security bookworm-security InRelease [34.8 kB]
Get:7 https://deb.debian.org/debian bookworm-backports/main Sources.diff/Index [63.3 kB]
Get:11 https://deb.debian.org/debian bookworm-backports/main Sources T-2026-09-14-1401.50-F-2026-09-14-1401.50.pdiff [299 B]
Get:11 https://deb.debian.org/debian bookworm-backports/main Sources T-2026-09-14-1401.50-F-2026-09-14-1401.50.pdiff [299 B]
Get:8 https://deb.debian.org/debian-security bookworm-security/main Sources [248 kB]
Get:9 https://deb.debian.org/debian-security bookworm-security/main amd64 Packages [338 kB]
Get:10 https://deb.debian.org/debian-security bookworm-security/main Translation-en [211 kB]
Fetched 1010 kB in 1s (870 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@localhost:~#
```

Les paquets sont à jour

3.7. Terraform destroy

```
user@debian:~/tform$ terraform destroy
terraform_data.upload_snippet: Refreshing state... [id=61d91797-848e-6914-2373-fc58]
macaddress.vmmac: Refreshing state... [id=e6:a6:fb:1c:85:04]
opnsense_kea_dhcpv4_reservation.debian_clone: Refreshing state... [id=926d685a-f8b29e-21d7df1c5490]
proxmox_vm_qemu.debian_clone: Refreshing state... [id=pve/qemu/120]
```

Terraform used the selected providers to generate the following execution plan. For actions are indicated with the following symbols:

- destroy

Terraform will perform the following actions:

```
# macaddress.vmmac will be destroyed
- resource "macaddress" "vmmac" {
  - address = "e6:a6:fb:1c:85:04" -> null
  - id      = "e6:a6:fb:1c:85:04" -> null
}

# opnsense_kea_dhcpv4_reservation.debian_clone will be destroyed
- resource "opnsense_kea_dhcpv4_reservation" "debian_clone" {
  - description = "Static IP for tf_cloned_debian" -> null
  - hostname    = "tf_cloned_debian" -> null
  - id          = "926d685a-f8ef-4996-b29e-21d7df1c5490" -> null
  - ip_address  = "10.0.0.140" -> null
  - mac_address = "e6:a6:fb:1c:85:04" -> null
  - subnet_id   = "4f275fc3-7f2d-4bdb-b8a7-eb794075aa7f" -> null
}

# proxmox_vm_qemu.debian_clone will be destroyed
- resource "proxmox_vm_qemu" "debian_clone" {
  - additional_wait = 5 -> null
  - agent           = 1 -> null
  - agent_timeout   = 90 -> null
  - automatic_reboot = true -> null
  - automatic_reboot_severity = "error" -> null
  - balloon         = 0 -> null
  - bios            = "seabios" -> null
  - boot            = "order=scsi0" -> null
  - cicustom        = "user=local:snippets/user-data.yml" -> null
  - ciupgrade       = false -> null
  - ciuser          = "admin" -> null
  - clone           = "debian" -> null
  - clone_wait      = 10 -> null
  - cores           = 0 -> null
  - current_node    = "pve" -> null
  - default_ipv4_address = "10.0.0.140" -> null
  - define_connection_info = true -> null
  - description     = "Managed by Terraform." -> null
}
```



```

- force_create          = false -> null
- full_clone            = true -> null
- hotplug               = "network,disk,usb" -> null
- id                    = "pve/qemu/120" -> null
- ipconfig0             = "ip=dhcp" -> null
- kvm                   = true -> null
- linked_vmid           = 0 -> null
- memory                = 2048 -> null
- name                  = "debian-TerraCloned" -> null
- numa                  = false -> null
- os_type               = "cloud-init" -> null
- protection            = false -> null
- qemu_os               = "l26" -> null
- reboot_required       = false -> null
- scsihw                = "lsi" -> null
- skip_ipv4             = false -> null
- skip_ipv6             = true -> null
- sockets               = 0 -> null
- ssh_host              = "10.0.0.140" -> null
- ssh_port              = "22" -> null
- sshkeys               = <<-EOT
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDcr42CrxxnGHxyUJQHGM4c0CQPjYyT0v
ydw user@debian
EOT -> null
- start_at_node_boot    = true -> null
- tablet                = true -> null
- tags                  = null
- target_node           = "pve" -> null
- unused_disk           = [] -> null
- vcpus                 = 0 -> null
- vm_state              = "running" -> null
- vmid                  = 120 -> null
# (9 unchanged attributes hidden)

- cpu {
-   cores      = 2 -> null
-   limit      = 0 -> null
-   numa       = false -> null
-   sockets    = 1 -> null
-   type       = "host" -> null
-   units      = 0 -> null
-   vcores     = 0 -> null
# (1 unchanged attribute hidden)
}

- disk {
-   backup      = true -> null
-   discard     = false -> null

```



```
- emulatesd           = false -> null
- format              = "raw" -> null
- id                  = 1 -> null
- iops_r_burst        = 0 -> null
- iops_r_burst_length = 0 -> null
- iops_r_concurrent   = 0 -> null
- iops_wr_burst       = 0 -> null
- iops_wr_burst_length = 0 -> null
- iops_wr_concurrent  = 0 -> null
- iothread            = false -> null
- linked_disk_id      = -1 -> null
- mbps_r_burst        = 0 -> null
- mbps_r_concurrent   = 0 -> null
- mbps_wr_burst       = 0 -> null
- mbps_wr_concurrent  = 0 -> null
- passthrough         = false -> null
- readonly            = false -> null
- replicate           = false -> null
- size                = "40G" -> null
- slot                = "scsi0" -> null
- storage              = "local-lvm" -> null
- type                = "disk" -> null
  # (6 unchanged attributes hidden)
}

- network {
  - bridge    = "vmb1" -> null
  - firewall  = false -> null
  - id        = 0 -> null
  - link_down = false -> null
  - macaddr   = "E6:A6:FB:1C:85:04" -> null
  - model     = "virtio" -> null
  - mtu       = 0 -> null
  - queues    = 0 -> null
  - rate      = 0 -> null
  - tag       = 0 -> null
}

- smbios {
  - uuid      = "58d80270-a06a-4187-9072-dc8e273c92fd" -> null
  # (6 unchanged attributes hidden)
}

- startup_shutdown {
  - order          = -1 -> null
  - shutdown_timeout = -1 -> null
  - startup_delay   = -1 -> null
}
```

```
    }  
  }  
  
# terraform_data.upload_snippet will be destroyed  
- resource "terraform_data" "upload_snippet" {  
  - id = "61d91797-848e-6914-2373-feb401291358" -> null  
  - triggers_replace = {  
    - snippet_hash = "76630081873a374637e8471c617d469e"  
  } -> null  
}
```

Plan: 0 to add, 0 to change, 4 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
opnsense_kea_dhcpv4_reservation.debian_clone: Destroying... [id=926d685a-f8ef-499  
1d7df1c5490]  
proxmox_vm_qemu.debian_clone: Destroying... [id=pve/qemu/120]  
opnsense_kea_dhcpv4_reservation.debian_clone: Destruction complete after 1s  
proxmox_vm_qemu.debian_clone: Destruction complete after 4s  
terraform_data.upload_snippet: Destroying... [id=61d91797-848e-6914-2373-feb40129  
macaddress.vmmac: Destroying... [id=e6:a6:fb:1c:85:04]  
terraform_data.upload_snippet: Destruction complete after 0s  
macaddress.vmmac: Destruction complete after 0s
```

Destroy complete! Resources: 4 destroyed.

user@debian:~/tform\$

4. Cleanup de l'espace de travail

Pour éviter de commiter des token, ranger correctement les variables et les providers, notamment pour travailler avec le versionnage git, il convient de ranger ces informations dans différents fichiers:

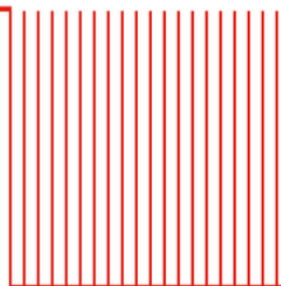
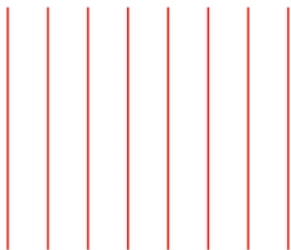
```
1 user@debian:~/terraform$ ls Shell  
2 main.tf provider.tf secrets.auto.tfvars terraform.tfstate user-data.yml.tpl var.tf  
3 user@debian:~/terraform$ cat .gitignore  
4 *.tfvars  
5 *.tfvars.json  
6  
7 **/.terraform/*  
8 *.tfstate  
9 *.tfstate.*  
10 crash.log  
11 crash.*.log  
12 *.tfplan  
13 user@debian:~/terraform$ cat provider.tf  
14 terraform {  
15   required_providers {  
16     proxmox = {
```

```
17     source = "telmate/proxmox"
18     version = "3.0.2-rc07" #3.0.2-rc07
19 }
20     macaddress = {
21         source = "ivoronin/macaddress"
22         version = "0.3.0"
23     }
24     opnsense = {
25         source = "browningluke/opnsense"
26         version = "~> 0.26.0"
27     }
28 }
29 }
30
31 provider "proxmox" {
32     pm_api_url          = var.pm_api_url
33     pm_api_token_id     = var.pm_api_token_id
34     pm_api_token_secret = var.pm_api_token_secret
35     pm_tls_insecure     = true
36     pm_minimum_permission_check = false
37 }
38
39 provider "opnsense" {
40     uri          = var.opnsense_uri
41     api_key      = var.opnsense_api_key
42     api_secret   = var.opnsense_api_secret
43     allow_insecure = true
44 }
45 user@debian:~/terraform$ cat secrets.auto.tfvars
46 pm_api_url          = "https://192.168.122.220:8006/api2/json"
47 pm_api_token_id     = "terraform@pve!tf-token"
48 pm_api_token_secret = "2ef5d121-2f70-48c7-a17b-78e7dc22d76e"
49
50 opnsense_uri        = "https://192.168.122.197"
51 opnsense_api_key    = "AncmTmUYXl5csabprgylFjDahbfRohVt7jYNm+UbKnPwAApQXaHXV7K3/
UeYLUkcFYlnCsRXYtnua/DC"
52 opnsense_api_secret =
"JU4/5kxzlfm5tALdiscA5Bztls6F+TZh6GoY2JLf0ITYUVadoVs4CjEq8lGH2kV1SNLaRLKQvVKxd370"
53 user@debian:~/terraform$ cat main.tf
54 locals {
55     ssh_public_key = trimspace(file(pathexpand(var.ssh_public_key_path)))
56 }
57
58 resource "terraform_data" "upload_snippet" {
59     triggers_replace = {
60         snippet_hash = md5(templatefile("${path.module}/user-data.yml.tpl", {
```

```
61     ci_user      = var.ci_user
62     ssh_public_key = local.ssh_public_key
63   )))
64 }
65
66 connection {
67   type      = "ssh"
68   user      = "root"
69   private_key = file(pathexpand("~/ssh/id_ed25519"))
70   host      = "192.168.122.220"
71 }
72
73 provisioner "remote-exec" {
74   inline = [
75     "mkdir -p /var/lib/vz/snippets/"
76   ]
77 }
78
79 provisioner "file" {
80   content = templatefile("${path.module}/user-data.yml.tpl", {
81     ci_user      = var.ci_user
82     ssh_public_key = local.ssh_public_key
83   })
84   destination = "/var/lib/vz/snippets/user-data.yml"
85 }
86 }
87
88 resource "macaddress" "vmmac" {
89   # prefix = [52, 54, 00]
90 }
91
92 resource "opnsense_kea_dhcpv4_reservation" "debian_clone" {
93   subnet_id   = "4f275fc3-7f2d-4bdb-b8a7-eb794075aa7f"
94   ip_address  = "10.0.0.140"
95   mac_address = lower(macaddress.vmmac.address)
96   hostname    = "tf_cloned_debian"
97   description = "Static IP for tf_cloned_debian"
98 }
99
100 resource "proxmox_vm_qemu" "debian_clone" {
101   name          = "debian-TerraCloned"
102   target_node   = "pve"
103
104   depends_on    = [terraform_data.upload_snippet]
105
106   os_type       = "cloud-init"
```

```
107   ciuser      = var.ci_user
108   sshkeys     = local.ssh_public_key
109   clone       = "debian"
110   full_clone  = true
111   agent       = 1
112   start_at_node_boot = true
113   # onboot = true
114   ipconfig0 = "ip=dhcp"
115   cicustom = "user=local:snippets/user-data.yml"
116
117   vmid        = 120
118   memory      = 2048
119   skip_ipv6   = true
120
121   cpu {
122     cores     = 2
123     sockets   = 1
124     type      = "host"
125   }
126
127   disk {
128     slot      = "scsi0" #ide2
129     size      = "40G"
130     type      = "disk" # cloudinit
131     storage   = "local-lvm"
132   }
133
134   network {
135     id        = 0
136     model     = "virtio"
137     bridge    = "vmbri1"
138     macaddr   = upper(macaddress.vmmac.address)
139   }
140 }
141 user@debian:~/terraform$ cat var.tf
142 variable "ci_user" {
143   description = "Utilisateur créé via cloud-init sur la VM"
144   type        = string
145   default     = "admin"
146 }
147
148 variable "ssh_public_key_path" {
149   description = "Chemin local vers la clé publique SSH à injecter dans la VM"
150   type        = string
151   default     = "~/ssh/id_ed25519.pub"
152 }
```

```
153
154 variable "pm_api_url" {
155     description = "URL de l'API Proxmox (ex: https://host:8006/api2/json)"
156     type        = string
157 }
158
159 variable "pm_api_token_id" {
160     description = "Identifiant du token API Proxmox (ex: terraform@pve!tf-token)"
161     type        = string
162 }
163
164 variable "pm_api_token_secret" {
165     description = "Secret du token API Proxmox"
166     type        = string
167     sensitive   = true
168 }
169
170 variable "opnsense_uri" {
171     description = "URL de l'API OPNsense"
172     type        = string
173 }
174
175 variable "opnsense_api_key" {
176     description = "Clé API OPNsense"
177     type        = string
178     sensitive   = true
179 }
180
181 variable "opnsense_api_secret" {
182     description = "Secret API OPNsense"
183     type        = string
184     sensitive   = true
185 }
```



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